

QMEA

1. A company wants to compare the mean productivity scores of employees who completed one of four different training programs. Different employees participated in each program. Which statistical test is most appropriate?

- A. Chi-Square Goodness-of-Fit Test
- B. Chi-Square Test of Independence
- C. One-Way ANOVA
- D. Randomized Block ANOVA

2. A One-Way ANOVA produces a statistically significant result. Which conclusion is justified?

- A. All population means are different from one another.
- B. At least one population mean differs from another.
- C. The population with the largest sample mean is significantly different from every other population.
- D. All population variances are different.

3. Four students each measure the same three objects. The researcher wants to determine whether the mean measurement error differs among the three objects, while accounting for differences among the students performing the measurements. Which test is most appropriate?

- A. One-Way ANOVA
- B. Randomized Block ANOVA
- C. Chi-Square Test of Independence
- D. Chi-Square Goodness-of-Fit Test

4. What is the main purpose of blocking in a Randomized Block ANOVA?

- A. To increase the number of treatments
- B. To convert quantitative data into categorical data
- C. To account for variation caused by a known nuisance factor
- D. To test whether two categorical variables are independent

5. A researcher compares three diets. Each of 10 participants tries all three diets, and a health outcome is measured after each diet. Why would a Randomized Block ANOVA be more appropriate than a One-Way ANOVA?

- A. There are exactly three treatments.
- B. The outcome is categorical.
- C. The observations across treatments are related through the same participants.
- D. The population variances are unknown.

6. A company claims that customers choose five product packages in proportions 10%, 20%, 30%, 20%, and 20%. A sample of customers is collected to determine whether the observed distribution agrees with these claimed proportions. Which test should be used?

- A. One-Way ANOVA
- B. Randomized Block ANOVA
- C. Chi-Square Goodness-of-Fit Test
- D. Chi-Square Test of Independence

7. A manufacturer records the work shift (Morning, Afternoon, Night) and shirt condition (Perfect, Flawed) for 600 shirts. The manager wants to know whether shirt condition is related to work shift. Which test should be used?

- A. One-Way ANOVA
- B. Chi-Square Goodness-of-Fit Test
- C. Chi-Square Test of Independence
- D. F-Test for Two Variances

8. Which situation should be analyzed using a Chi-Square Goodness-of-Fit Test rather than a Chi-Square Test of Independence?

- A. Determine whether gender is associated with preferred payment method.
- B. Determine whether shirt quality is associated with work shift.
- C. Determine whether customer choices among four brands follow proportions of 20%, 30%, 30%, and 20%.
- D. Determine whether education level is associated with employment status.

9. A researcher wants to determine whether job satisfaction (Satisfied/Not Satisfied) differs among employees in four departments. Which test is most appropriate?

- A. One-Way ANOVA
- B. Randomized Block ANOVA
- C. Chi-Square Goodness-of-Fit Test
- D. Chi-Square Test of Independence

10. A company wants to compare the effectiveness of three employee training programs. Thirty employees are randomly assigned so that 10 different employees receive each program. After training, each employee takes the same productivity test.

Which statistical test is most appropriate?

- A. Randomized Block ANOVA
- B. One-Way ANOVA
- C. Chi-Square Goodness-of-Fit Test
- D. Chi-Square Test of Independence

11. A professor wants to compare three teaching methods. Instead of assigning different students to each method, each of 20 students completes a learning task using all three methods. The professor wants to account for differences in ability among students.

Which test is most appropriate?

- A. One-Way ANOVA
- B. Randomized Block ANOVA
- C. Chi-Square Test of Independence
- D. Chi-Square Goodness-of-Fit Test

12. Four technicians each inspect products produced by three machines. Every technician evaluates products from all three machines. The researcher wants to determine whether mean inspection scores differ among machines while controlling for differences among technicians.

In a Randomized Block ANOVA, what are the treatments and blocks?

- A. Treatments = technicians; Blocks = machines
- B. Treatments = machines; Blocks = technicians
- C. Treatments = inspection scores; Blocks = machines
- D. Treatments = products; Blocks = inspection scores

13. A researcher measures customer satisfaction in three branches.

Study A: Satisfaction is recorded as a score from 0–100.

Study B: Satisfaction is recorded as Satisfied / Not Satisfied.

Different customers are sampled at each branch.

Which pair of tests is most appropriate?

- A. Study A: One-Way ANOVA; Study B: Chi-Square Test of Independence
- B. Study A: Chi-Square Independence; Study B: One-Way ANOVA
- C. Study A: Randomized Block ANOVA; Study B: Goodness-of-Fit
- D. Both studies: One-Way ANOVA

14. A supermarket wants to study customers' preferred payment method.

Study 1: It tests whether Cash/Card/QR payments occur in expected proportions of 20%, 30%, and 50%.

Study 2: It tests whether payment method is associated with age group (Under 30 / 30–49 / 50+).

Which tests should be used?

- A. Study 1: Independence; Study 2: Goodness-of-Fit
- B. Study 1: Goodness-of-Fit; Study 2: Independence
- C. Study 1: One-Way ANOVA; Study 2: Independence
- D. Both: Goodness-of-Fit

15. A researcher wants to compare the average waiting time at three hospitals. She samples 50 different patients from each hospital. Another researcher measures waiting time for the same 30 patients under three different scheduling systems.

Which tests are most appropriate?

- A. Both studies: One-Way ANOVA
- B. First: Randomized Block ANOVA; Second: One-Way ANOVA
- C. First: One-Way ANOVA; Second: Randomized Block ANOVA
- D. First: Chi-Square Independence; Second: Randomized Block ANOVA